

Lecture 8 — Mathematics of Projection

Lecture focus: How 3D objects are projected onto a 2D view plane. This lecture explains projection taxonomy, parallel projection, orthographic projection, oblique projection, axonometric/isometric projection, perspective projection, vanishing points, and the basic mathematics behind projection matrices.

Learning outcomes

1. Explain why projection is required in 3D computer graphics.
 2. Distinguish between **parallel projection** and **perspective projection**.
 3. Explain orthographic, oblique, axonometric, and isometric projection.
 4. Explain the pinhole camera model of perspective projection.
 5. Derive the basic perspective projection formula using similar triangles.
 6. Explain why distant objects appear smaller in perspective projection.
 7. Explain vanishing points and one-, two-, and three-point perspective.
 8. Understand the conceptual role of projection matrices.
 9. Explain why depth information is still needed after projection.
 10. Relate projection to games, CAD, photography, architecture, robotics, maps, and scientific visualization.
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1. Why projection is necessary

A 3D object has three coordinates:

$$P = (x, y, z)$$

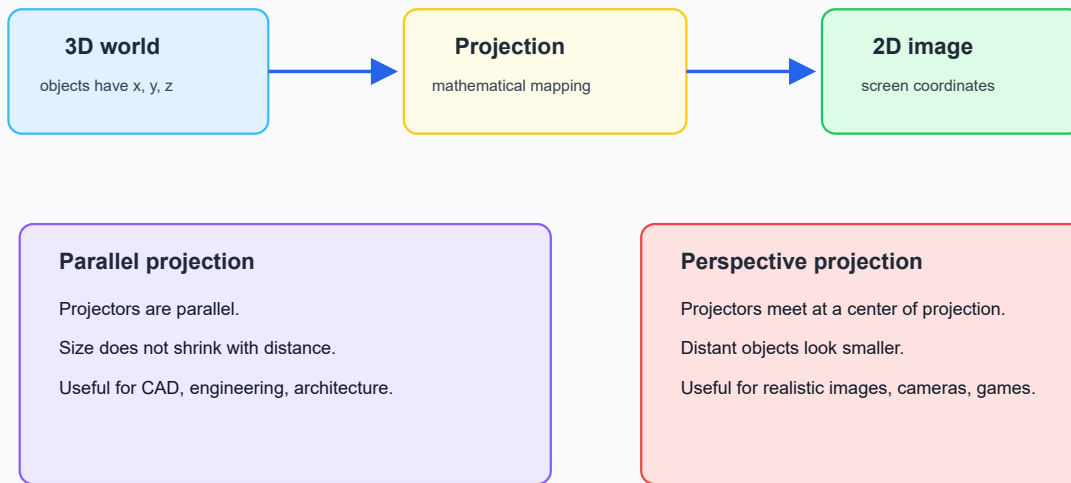
A screen, paper, or image has only two main spatial coordinates:

$$P' = (x', y')$$

So a graphics system needs a method for converting a 3D scene into a 2D image.

That method is called **projection**.

Projection: Turning 3D Geometry into a 2D Image



Projection answers a fundamental question:

Where should a 3D point appear on the 2D screen?

In a full graphics pipeline, projection usually happens after modeling and viewing transformations:

Model transformation → View transformation → Projection transformation → Screen mapping

A simple analogy

Your eye sees a 3D world, but the image formed on the retina is 2D. A camera photographs a 3D world, but the photograph is 2D. A computer graphics system does something similar using mathematics.

2. Projection as a mapping

Mathematically, projection maps a 3D point to a 2D point:

$$(x, y, z) \rightarrow (x', y')$$

Different projection methods produce different visual results.

Some projections preserve measurement better. Some projections create realistic depth. Some projections are easier to draw. Some projections are better for engineering and architecture.

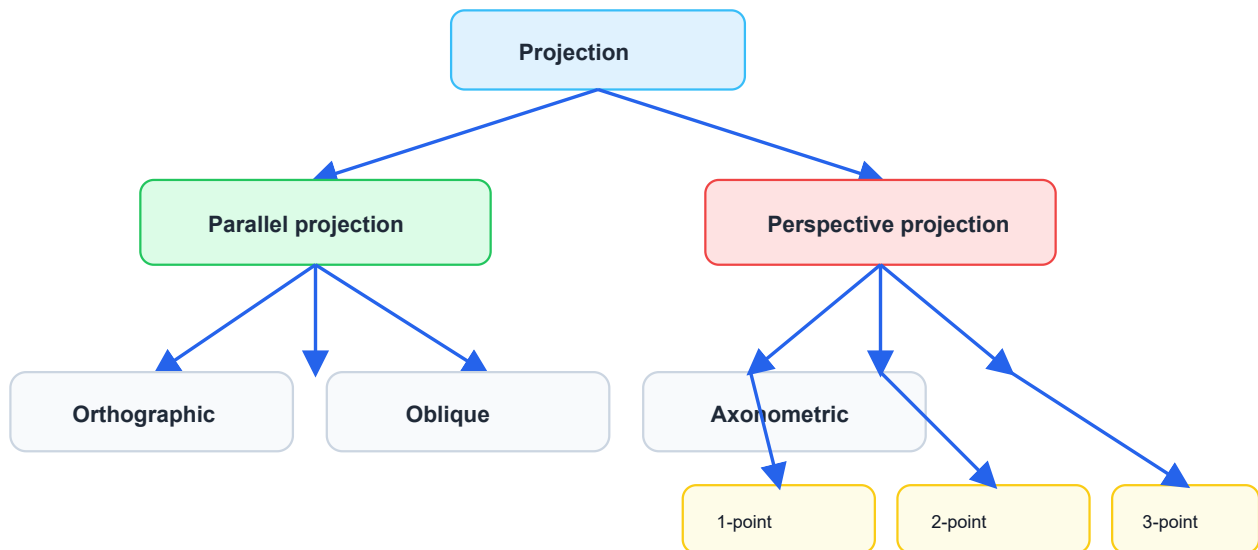
This is why projection is not only a mathematical topic but also a design choice.

3. Taxonomy of projection

Projection is commonly divided into two major families:

1. Parallel projection
2. Perspective projection

Taxonomy of Projection



Parallel projection

In parallel projection, projection lines are parallel to one another.

Important subtypes:

- orthographic projection,
- oblique projection,
- axonometric projection,
- isometric projection.

Perspective projection

In perspective projection, projection lines meet at a single point called the **center of projection**.

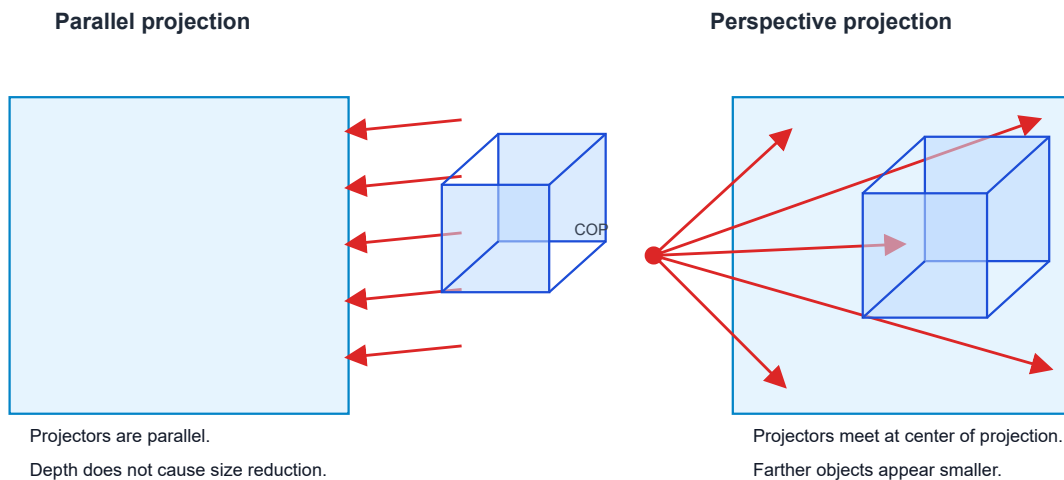
Important subtypes:

- one-point perspective,
- two-point perspective,
- three-point perspective.

4. Parallel projection vs perspective projection

The main difference is the direction of projection lines.

Parallel vs Perspective Projectors



Parallel projection

In parallel projection:

projection lines are parallel

This means an object does not appear smaller just because it is farther away.

Parallel projection is useful when accurate shape and measurement are more important than visual realism.

Perspective projection

In perspective projection:

projection lines meet at a center of projection

This makes distant objects appear smaller.

Perspective projection is closer to how human vision and cameras behave.

Quick comparison

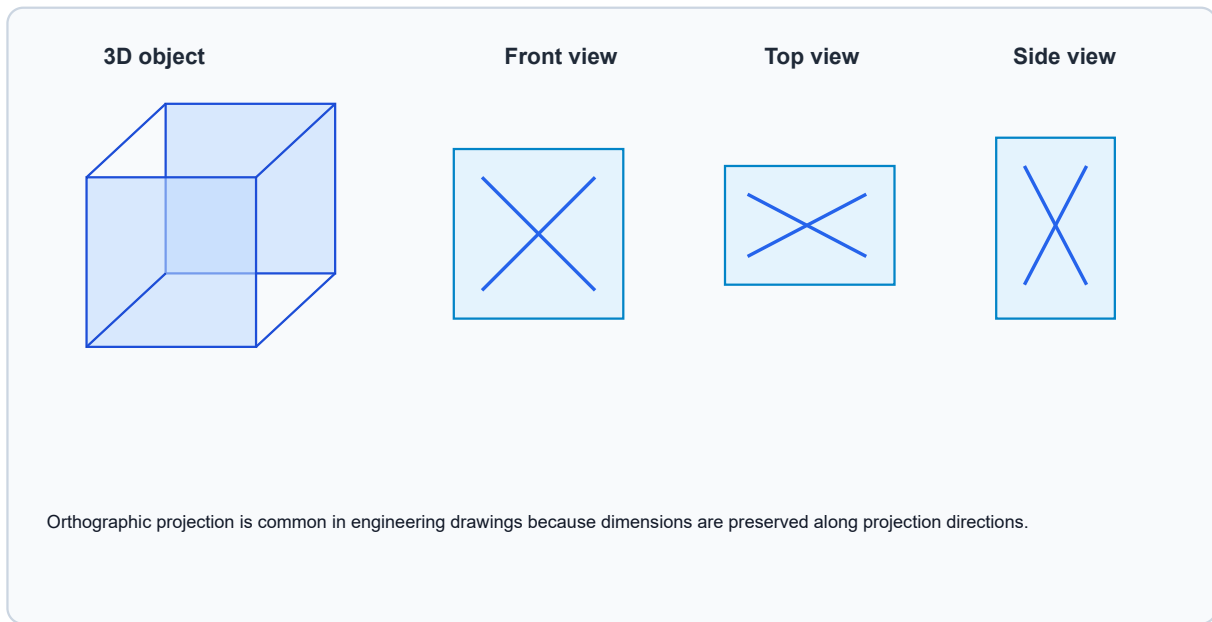
Feature	Parallel projection	Perspective projection
Projection lines	Parallel	Meet at one point
Distant objects	Same size	Appear smaller
Realism	Less realistic	More realistic
Measurement	Easier	Distorted by depth
Common use	CAD, engineering	Games, films, visualization

5. Orthographic projection

Orthographic projection is a type of parallel projection where the projection lines are perpendicular to the view plane.

It is widely used in engineering and technical drawings.

Orthographic Projection: Front, Top, and Side Views



Common orthographic views include:

- front view,
- top view,
- side view.

Why orthographic projection is useful

Orthographic projection does not shrink objects with distance. This makes it useful for precise measurement.

For example, if a mechanical part is 10 cm wide, an orthographic drawing can show that width consistently without perspective distortion.

Application

Orthographic projection is used in:

- engineering drawings,
- architectural plans,
- CAD software,
- product design,
- machine part diagrams,
- technical manuals.

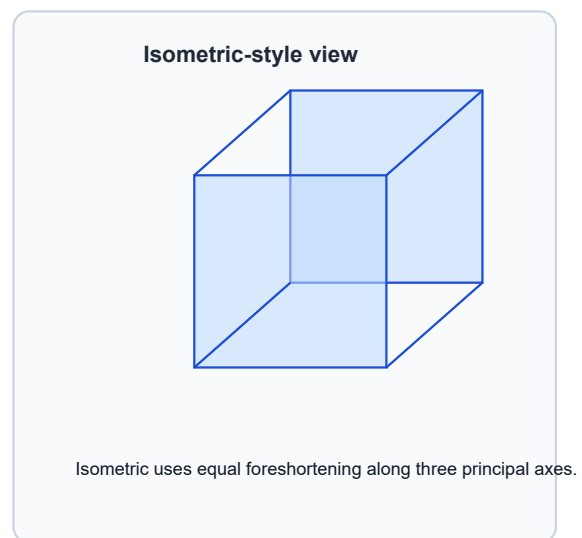
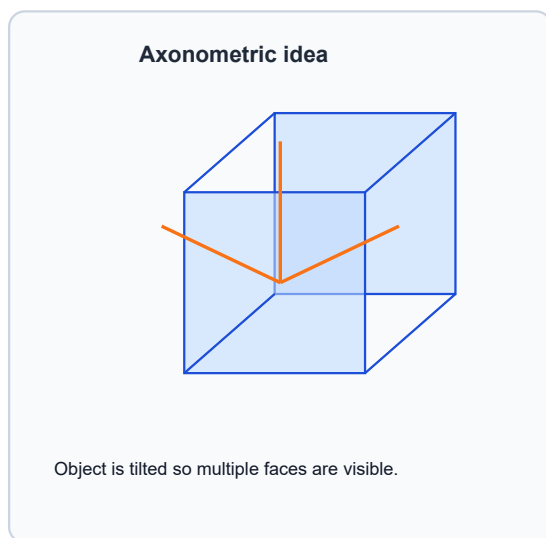
Trivia

Many engineering drawings use multiple orthographic views together because one view alone cannot fully describe a 3D object. A front view may hide depth, while a top or side view reveals it.

6. Axonometric and isometric projection

Axonometric projection is a type of parallel projection where the object is rotated relative to the projection plane so that multiple faces are visible.

Axonometric Projection and the Isometric View



A common special case is **isometric projection**.

In isometric projection, the three principal axes are equally foreshortened. This gives a balanced 3D-like view without perspective shrinking.

Why isometric projection is popular

Isometric projection is visually clear and easy to understand.

It is used in:

- technical illustrations,
- game art,
- diagrams,

- assembly instructions,
- architectural concept drawings.

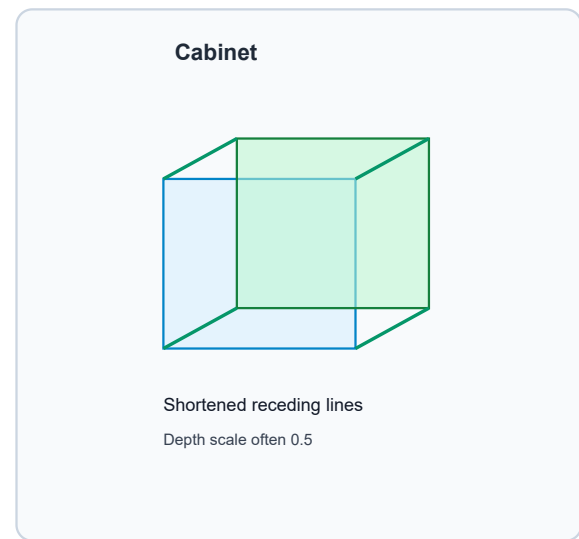
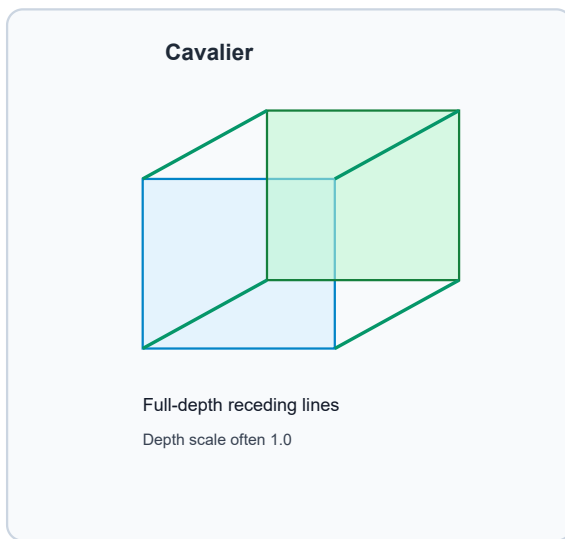
Trivia

Many classic strategy games and simulation games use isometric or pseudo-isometric views. They look 3D-like but are often easier to render and control than full perspective 3D.

7. Oblique projection

In **oblique projection**, the front face of an object is shown true to shape, while depth is represented using angled receding lines.

Oblique Projection: Cavalier and Cabinet Ideas



Two common forms are:

Cavalier projection

In cavalier projection, the receding depth is often drawn at full scale.

This can make the object look deep or stretched.

Cabinet projection

In cabinet projection, the receding depth is usually shortened, often by half.

This tends to look more natural.

Usefulness

Oblique projection is easy to draw by hand and can clearly show the front face of an object.

It is useful in simple technical sketches, textbook diagrams, and explanatory illustrations.

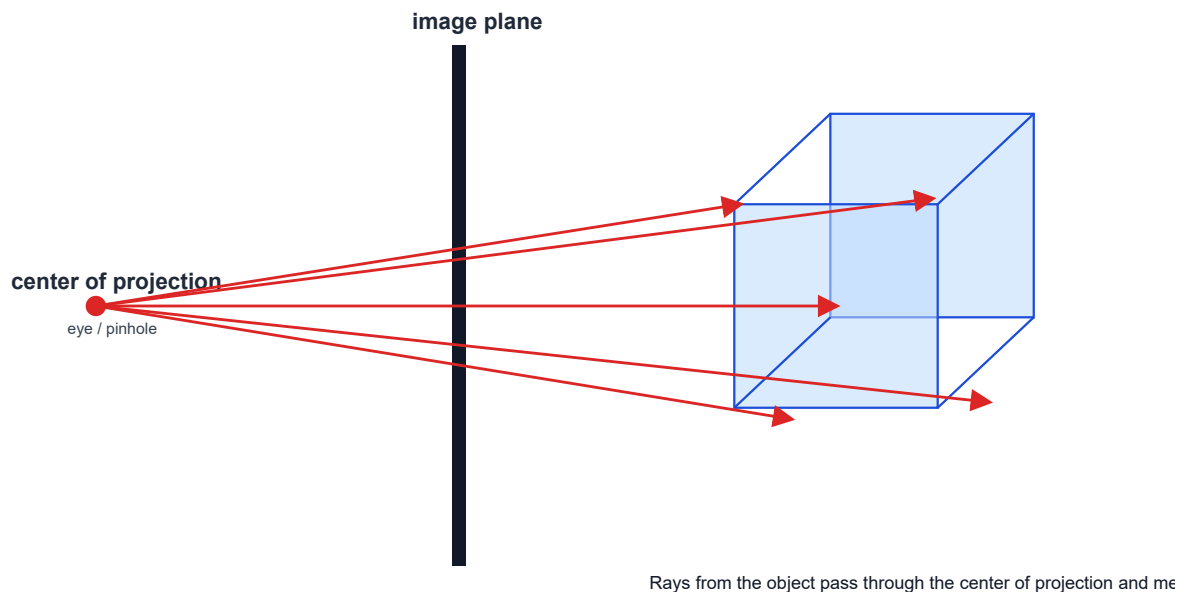
8. Perspective projection

Perspective projection is the projection method that gives the strongest sense of realism.

It is based on the idea that projection rays pass through a common point called the **center of projection**.

This is similar to a pinhole camera.

Perspective Projection: Pinhole Camera Model



Key terms

Center of projection, or COP:

The point through which projection rays pass. In a camera analogy, this is the pinhole or eye.

Projection plane:

The plane where the image is formed.

Projectors:

The lines connecting 3D points to the center of projection.

Main visual effect

Objects farther away appear smaller.

That single effect is responsible for much of the realism of perspective images.

9. Perspective formula from similar triangles

Assume the center of projection is at the origin and the projection plane is at:

$$z = d$$

A 3D point is:

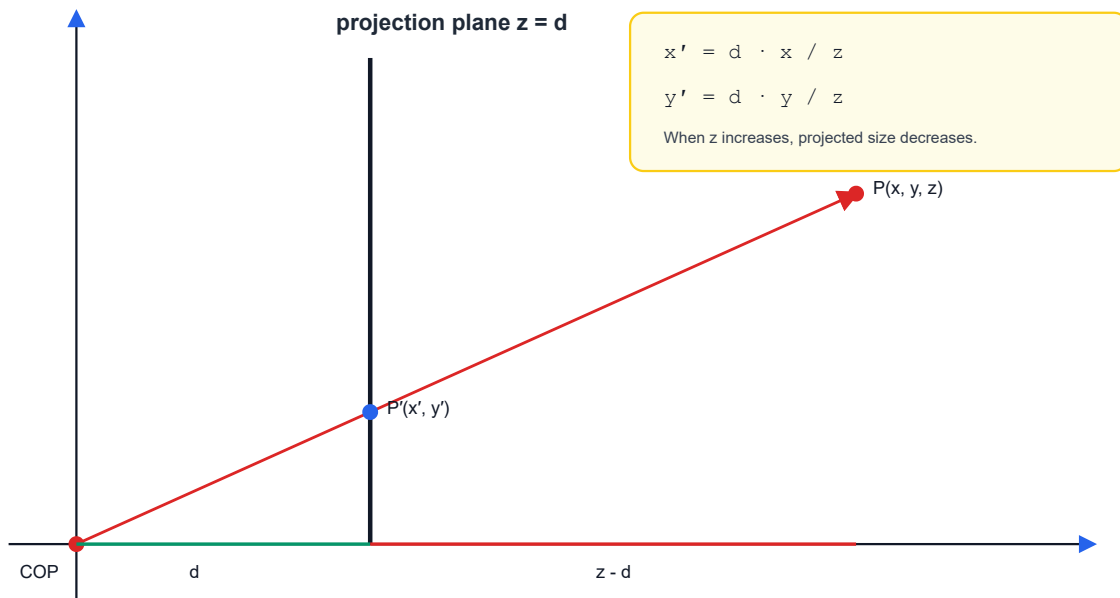
$$P = (x, y, z)$$

The projected point is:

$$P' = (x', y')$$

Using similar triangles:

Perspective Formula from Similar Triangles



The basic perspective projection equations are:

$$x' = d \cdot x / z$$
$$y' = d \cdot y / z$$

What the formula means

If z becomes larger, then:

x' and y' become smaller

So the object appears closer to the center of the image.

This is why distant objects look smaller.

Example

Let:

$$d = 2$$
$$P = (4, 6, 8)$$

Then:

$$x' = 2 \times 4 / 8 = 1$$
$$y' = 2 \times 6 / 8 = 1.5$$

So the projected point is:

$$P' = (1, 1.5)$$

Now if the same x and y values are farther away:

$$P = (4, 6, 16)$$

then:

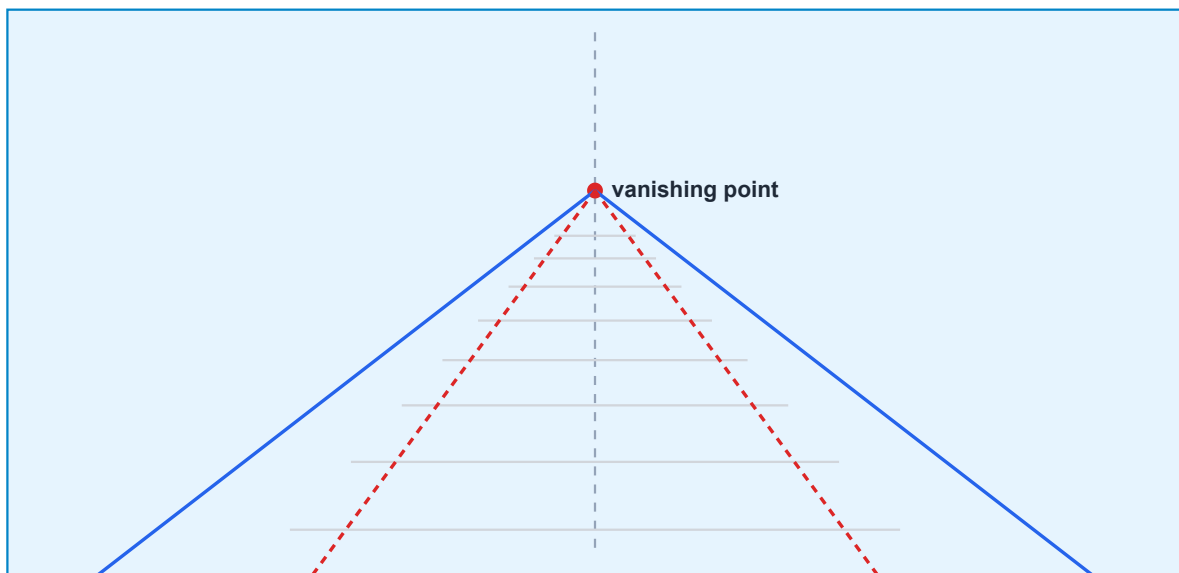
$$x' = 2 \times 4 / 16 = 0.5$$
$$y' = 2 \times 6 / 16 = 0.75$$

The projected point is closer to the center, so the object appears smaller.

10. Vanishing points

A **vanishing point** is a point in the image where parallel lines in 3D appear to meet.

Vanishing Point: Parallel Lines Meeting in the Image



Parallel 3D lines that move away from the viewer appear to converge at a vanishing point.

In the real world, railway tracks are parallel. But in a photograph, they appear to converge in the distance.

This happens because of perspective projection.

Why vanishing points matter

Vanishing points are important in:

- drawing,
- photography,
- architecture,
- camera calibration,
- 3D reconstruction,
- visual storytelling.

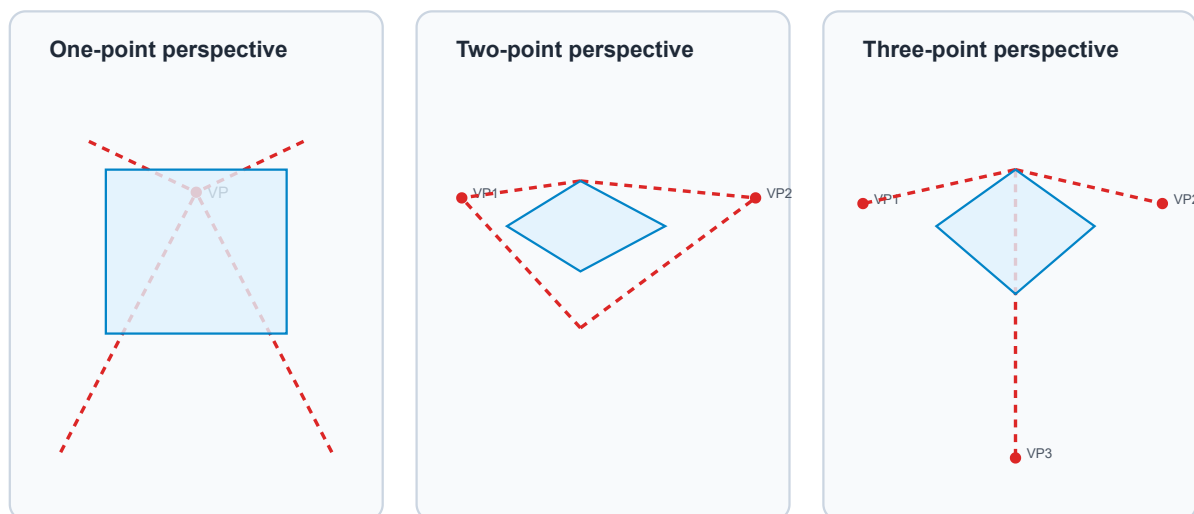
Important note

Parallel lines only appear to meet in the projected image. In the actual 3D world, they remain parallel.

11. One-, two-, and three-point perspective

Perspective drawing is often classified by how many principal vanishing points are visible.

One-, Two-, and Three-Point Perspective



More tilted object/camera orientations produce more vanishing directions.

One-point perspective

One-point perspective occurs when one major direction of the object goes into depth.

Example:

- looking straight down a road,
- looking straight into a corridor,
- a front-facing box with depth lines receding.

Two-point perspective

Two-point perspective occurs when two major horizontal directions recede to two different vanishing points.

Example:

- viewing the corner of a building.

Three-point perspective

Three-point perspective occurs when vertical lines also converge.

Example:

- looking up at a tall building,
- looking down from a high height.

12. Projection matrices: conceptual view

In modern graphics pipelines, projection is usually represented using matrices.

Projection Matrices: Conceptual View

Orthographic projection

$$\begin{aligned} \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \end{aligned}$$

Drops or ignores depth for image location.

No perspective divide.

Perspective projection

$$\begin{aligned} x_{\text{clip}} &= d \cdot x \\ y_{\text{clip}} &= d \cdot y \\ w_{\text{clip}} &= z \\ x_{\text{screen}} &= x_{\text{clip}} / w_{\text{clip}} \\ y_{\text{screen}} &= y_{\text{clip}} / w_{\text{clip}} \end{aligned}$$

The divide by z creates perspective shrinking.

Orthographic projection matrix idea

Orthographic projection can be understood as dropping the depth coordinate for image placement:

$$(x, y, z) \rightarrow (x, y)$$

Depth may still be stored separately for visibility testing.

Perspective projection matrix idea

Perspective projection needs division by depth:

$$\begin{aligned} x' &= d \cdot x / z \\ y' &= d \cdot y / z \end{aligned}$$

Using homogeneous coordinates, the projection matrix prepares values so that after matrix multiplication, the graphics pipeline performs a **perspective divide**:

$$\begin{aligned} x_{\text{screen}} &= x_{\text{clip}} / w_{\text{clip}} \\ y_{\text{screen}} &= y_{\text{clip}} / w_{\text{clip}} \\ z_{\text{screen}} &= z_{\text{clip}} / w_{\text{clip}} \end{aligned}$$

For simple perspective:

`w_clip` is related to `z`

This division is what creates perspective shrinking.

Why homogeneous coordinates are useful here

Projection is one of the strongest reasons homogeneous coordinates are used in computer graphics.

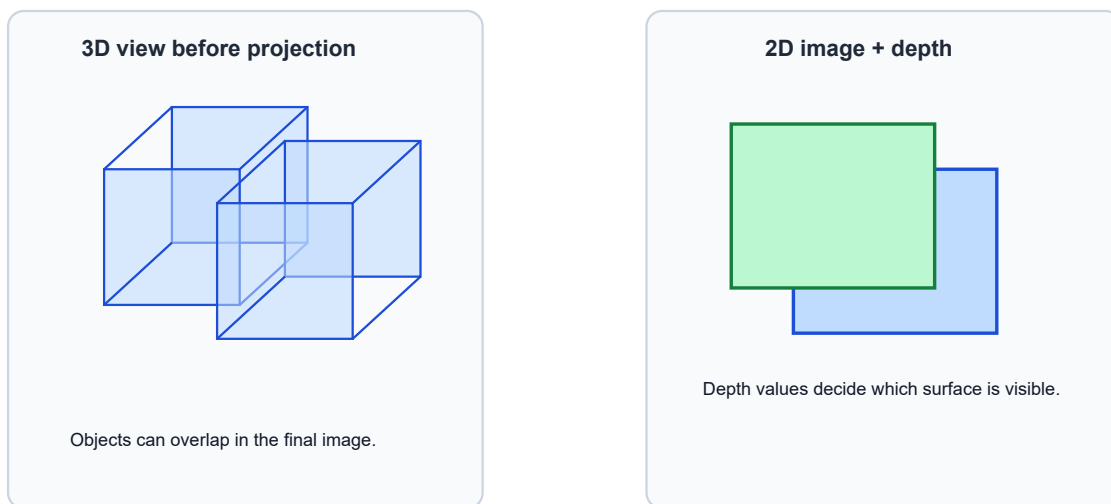
They allow perspective projection to be represented within a matrix pipeline.

13. Projection and depth

Projection maps 3D points to 2D positions. But after projection, different 3D points may map to the same screen position.

So projection alone does not solve visibility.

Projection Does Not Remove the Need for Depth



Objects can overlap in the final image.

Depth values decide which surface is visible.

Projection gives screen position; hidden-surface methods decide visibility.

A graphics system still needs depth information to decide which surface is visible.

This is why later topics such as **hidden surface removal** and the **Z-buffer algorithm** are important.

Key idea

Projection answers:

Where does the point appear on the screen?

Depth testing answers:

Which point is visible if multiple points project to the same screen area?

14. Projection choices in real applications

Applications of Projection

Games and real-time graphics

Perspective cameras create a believable 3D world.

CAD and engineering

Orthographic views preserve measurements and shape.

Architecture

Perspective previews plus technical drawings.

Photography and vision

Pinhole camera model is a geometric foundation.

Maps and GIS

Parallel projection is useful for measurements and layers.

Scientific visualization

Choose views that reveal structure in 3D data.

Choosing the projection is a design decision: realism, measurement accuracy, or clarity may be more important depending on the task.

Different applications use different projection methods.

Games

Games usually use perspective projection for the main camera because it looks realistic.

CAD and engineering

CAD software often uses orthographic projection because measurement accuracy is important.

Architecture

Architecture may use both:

- orthographic projection for plans and elevations,
- perspective projection for realistic previews.

Photography and computer vision

The pinhole camera model is a foundational approximation for camera geometry.

Maps and GIS

Parallel projection is useful when distances and shapes need controlled representation.

Scientific visualization

Researchers choose projections that best reveal the structure of data.

15. Common student mistakes

1. Thinking projection and viewing transformation are the same thing.
 2. Forgetting that projection converts 3D to 2D.
 3. Saying perspective projection uses parallel projectors.
 4. Thinking parallel lines actually meet in 3D.
 5. Forgetting that orthographic projection does not shrink objects with distance.
 6. Confusing oblique projection with perspective projection.
 7. Forgetting that perspective projection involves division by depth.
 8. Assuming projection alone solves hidden surface visibility.
 9. Mixing up center of projection and projection plane.
 10. Ignoring coordinate-system conventions when writing projection formulas.
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16. Worked examples

Example 1: Orthographic projection

Project:

$$P = (4, 7, 10)$$

onto the xy-plane using simple orthographic projection.

Since orthographic projection ignores z for image position:

$$P' = (4, 7)$$

Depth may still be stored separately as:

$$z = 10$$

Example 2: Perspective projection

Let the projection plane be:

$$z = d = 5$$

Project:

$$P = (10, 15, 25)$$

Using:

$$\begin{aligned}x' &= d \cdot x / z \\y' &= d \cdot y / z\end{aligned}$$

we get:

$$\begin{aligned}x' &= 5 \times 10 / 25 = 2 \\y' &= 5 \times 15 / 25 = 3\end{aligned}$$

So:

$$P' = (2, 3)$$

Example 3: Effect of distance

Let:

$$d = 2$$

Compare two points:

$$A = (4, 4, 4)$$

$$B = (4, 4, 8)$$

For A:

$$x' = 2 \times 4 / 4 = 2$$

$$y' = 2 \times 4 / 4 = 2$$

For B:

$$x' = 2 \times 4 / 8 = 1$$

$$y' = 2 \times 4 / 8 = 1$$

Although A and B have the same x and y values, B is farther away, so its projection is closer to the center.

17. Short exam-style questions

1. What is projection in computer graphics?
 2. Why is projection needed in 3D graphics?
 3. What is the difference between parallel and perspective projection?
 4. What is orthographic projection?
 5. Why is orthographic projection useful in engineering drawings?
 6. What is oblique projection?
 7. What is the difference between cavalier and cabinet projection?
 8. What is the center of projection?
 9. Derive the basic perspective formulas $x' = dx/z$ and $y' = dy/z$.
 10. What is a vanishing point?
 11. What is the difference between one-point and two-point perspective?
 12. Why is depth still needed after projection?
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18. Practice problems

Problem 1

Using simple orthographic projection onto the xy-plane, project:

$$P = (8, 3, 12)$$

Problem 2

Using perspective projection with:

$$d = 4$$

project:

$$P = (12, 8, 16)$$

Problem 3

Two points are given:

$$A = (6, 6, 6)$$

$$B = (6, 6, 18)$$

Using perspective projection with $d = 3$, compute their projected coordinates. Which appears closer to the image center?

Problem 4

Explain why a circle may appear as an ellipse under some projection/viewing conditions.

Problem 5

Name one application where orthographic projection is better than perspective projection and explain why.

19. Lecture summary

Lecture 8 Summary

- | | |
|----------------------------------|---|
| ● Projection | maps 3D points to a 2D image. |
| ● Parallel projection | uses parallel projectors and preserves relative size. |
| ● Orthographic projection | is useful for technical drawings and CAD. |
| ● Oblique projection | shows depth using angled projectors. |
| ● Perspective projection | uses a center of projection and creates realistic shrinking. |
| ● Vanishing points | appear where parallel 3D lines seem to meet. |
| ● Projection matrices | use homogeneous coordinates and sometimes perspective divide. |
| ● Depth | is still needed to decide visibility after projection. |

Big idea: projection is the mathematical camera of computer graphics.

The big takeaways are:

- Projection maps 3D points onto a 2D image.
- Parallel projection uses parallel projectors.
- Perspective projection uses a center of projection.
- Orthographic projection is useful for technical and engineering drawings.
- Oblique projection shows depth using angled projectors.
- Axonometric and isometric projections show multiple faces without perspective shrinking.
- Perspective projection creates realistic size reduction with distance.
- Similar triangles give the basic formulas $x' = dx/z$ and $y' = dy/z$.
- Vanishing points are a natural result of perspective projection.
- Projection matrices are central to modern graphics pipelines.
- Projection gives screen position, but depth testing is still needed for visibility.

20. Final trivia box

- The pinhole camera model is simple, but it is one of the most important ideas in computer vision and graphics.
- Renaissance artists studied perspective geometry centuries before computer graphics existed.
- CAD programs often let users instantly switch between orthographic and perspective views.
- A game camera is essentially a moving projection system.

- The perspective divide is one of the most important steps in the GPU graphics pipeline.